

		answer
4(a)	The internal energy of a gas is the sum of a random distribution of kinetic and potential energies associated with the molecules of the gas.	[1]
4(b)(i)	$N = \frac{(3.3 \times 10^5)(1.8 \times 10^{-3})}{(1.38 \times 10^{-23})(37 + 273.15)}$ $N = 1.39 \times 10^{23} \text{ molecules}$ $N \approx 1.4 \times 10^{23} \text{ molecules}$	[1] for correct substitution and showing 1.39×10^{23}
4(b)(ii)1	$\Delta U = \frac{3}{2} Nk\Delta T$ $\Delta U = \frac{3}{2} (1.4 \times 10^{23})(1.38 \times 10^{-23})(15 - 37)$ $\Delta U = -63.756 \text{ J}$ $\Delta U \approx -64 \text{ J}$ (Accept -63 J for using $3/2 pV$)	[1] for correct substitution [1] for correct answer Minus one mark if there is no negative" sign

4(b)(ii)2	$\Delta U = Q + W$ $-64 = Q - 85$ $Q = 21 \text{ J}$	[1] for correct substitution [1] for correct answer
4(b)(ii)3	r.m.s. speed $\propto \sqrt{T}$ Ratio $= \frac{\sqrt{15 + 273.15}}{\sqrt{37 + 273.15}}$ $= 0.9639$ ≈ 0.96	[1] for correct substitution [1] for correct answer Allow other method.
5(a)	Progressive: There is a net transmission of energy from the	[1]